

Assignment-03

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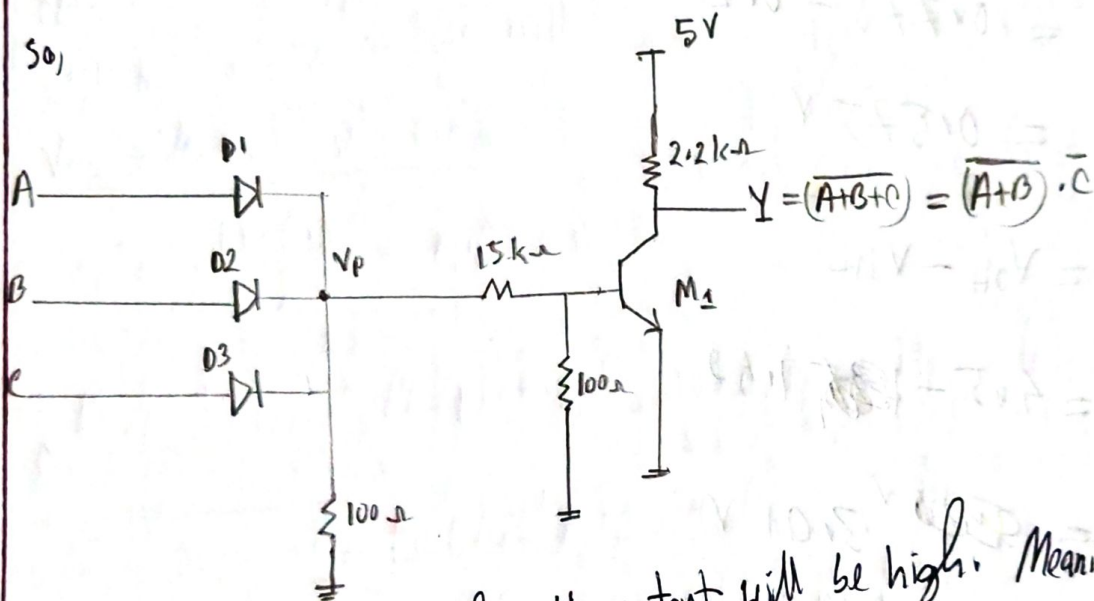
Section: 15

Ans to the Q No-1

$$Y = \overline{A+B} \cdot \bar{C}$$

$$= \overline{(A+B+C)}$$

$$[\because (A+B)' = A' \cdot B']$$

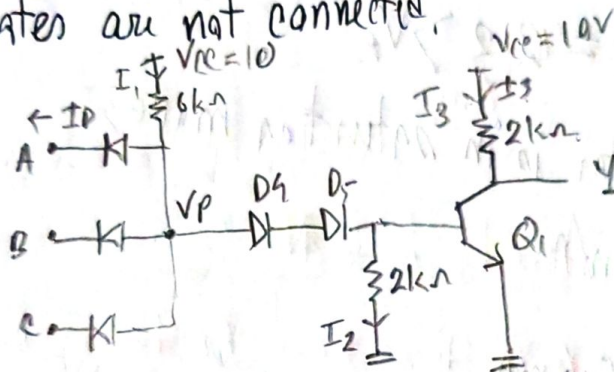


If only A, B, C all are low the output will be high. Meaning if A, B, C are low, D1, D2, D3 diodes are off V_p is also low meaning M_1 will be in cut-off mode. So Y output will obviously be high.

(Ans)

Ans to the Q No-2

If load gates are not connected.



$$\text{Diode cut in} = 0.6V$$

$$\text{Transistor cut in} = 0.5V$$

To get the transistor on $V_p = 0.6 + 0.6 + 0.5 = 1.7V$.

(a) If any input is low, $V_A = 0.1V$.

then,

$$V_p' = (0.7 + 0.1)V = 0.8V < V_p$$

So Q_1 will be cut off.

$$I_2 = I_{E1} = I_{B1} = 0mA$$

Now,

$$I_1 = \frac{10 - V_0}{6}$$

$$I_1 = \frac{10 - 0.8}{6}$$

$$I_1 = 1.533mA$$

$$\therefore I_0 = \cancel{0.82mA} \quad I_1 = 1.533$$

$$\therefore \text{Power dissipation} = (10 - 0.1) \times I_1$$

$$= 15.176 \text{ mW}$$

if two inputs were low,

$$i_D = \frac{I_1}{2} = \frac{1.533}{2} = 0.76mA$$

if all inputs were low,

$$i_D = \frac{I_1}{3} = \frac{1.533}{3} = 0.511mA$$

(Ans)

(b) If all inputs were high -

$$V_r' = 0.7 + 0.7 + 0.8 = 2.2 > V_p$$

So the Q2 transistor is in saturation mode;

$$I_1 = \frac{10 - 2.2}{6} = 1.3 \text{ mA}$$

$$I_2 = \frac{0.8 - 0}{2} = 0.4 \text{ mA}$$

$$I_{B1} = I_1 - I_2 = 0.9 \text{ mA}$$

$$V_{CE} = 0.2 \text{ V}, V_e = 0.2 \text{ V}$$

$$I_{C1} = \frac{10 - 0.2}{2} = 4.95 \text{ mA}$$

$$\therefore \text{Power dissipation} = (10 - 0)i_2 + (10 - 0)i_{B1} + (10 - 0)i_{C1}$$

$$= (10 - 0)(i_2 + i_{B1} + i_{C1})$$

$$= 10(0.4 + 0.9 + 4.95) \text{ mW}$$

$$= 10 \times 6.25$$

$$= 62.5 \text{ mW}$$

Ans)

(C) As ~~load~~ load gates are now connected to the input driver, so there will be two cases, $V = \text{low}$ or $V = \text{high}$.

Case I : $V = \text{high}$, $V = 10\text{V}$.

D_0 will be off so no current will enter ~~load gates~~ ^{load gates}, so supply and demand current $= 0\text{mA}$.

$$\therefore I_{\text{demand}} = 0$$

Case II $V = \text{low}$, $V = 0.1\text{V}$.

$$I_L = \frac{10 - 0.7 - 0.1}{6} = 1.533\text{mA}$$

$$I_{E1} = \frac{10 - 0.1}{2} = 4.95\text{mA}$$

$$I_{B1} = 0.9\text{mA}$$

$$\text{So } \beta_F = \frac{I_{C\text{max}}}{I_{B1}}$$

$$\therefore I_{C\text{max}} = 30 \times 0.9 = 27\text{mA}$$

We know,

$$I_{C\text{max}} = I_E + N \times I_L$$

$$27 = 4.95 + N \times 1.533$$

$$N = \frac{27 - 4.95}{1.533} = 14.38$$

$$\therefore N = 14$$

$$\therefore \text{Maximum } I_{\text{out}} = \min(\infty, 14) \\ = 14.$$

Ans)

Ans to the Q No-3

(a)

As $V_{in} = \text{high}$, $V_{CE} = 0.2 \text{ V}$, $R_{in} = 5 \text{ k}\Omega$, $R_D = 100 \text{ k}\Omega$, $R_C = 2 \text{ k}\Omega$, $n = 48$

Now,

$$I_{R_C} = \frac{5 - 0.2}{2} = 2.4 \text{ mA}$$

$$I_{R_B} = \frac{0.8 + 5}{100} = 0.058 \text{ mA} \quad [\because V_B = 0.8]$$

$$I_1 = \frac{5 - 0.8}{5} = 0.84 \text{ mA}$$

$$I_{B_1} = I_1 - I_{R_B} = 0.782 \text{ mA}$$

Now

$$P = (V_{CE} - 0) \times I_{R_C} + (V_{in} - 0) \cdot I_{B_1} + (V_{in} + 5) i_{R_D}$$

$$= (5 - 0) \times \left(I_{R_C} - 48 \times \frac{0.2}{105} \right) + (5 - 0) \times 0.782 + 10 (0.058) \text{ mW}$$

$$= 5 \left(2.4 - 48 \times \frac{0.2}{105} \right) + 5 \times 0.782 + 10 \times 0.058 \text{ mW}$$

$$= (11.54 + 3.91 + 0.58) \text{ mW}$$

$$= 16.03 \text{ mW}$$

Ans

b) $V_{OH} = 4.5V$, $V_{OL} = 0.2V$.

Determination of V_{IL} ,

Transistor is on the verge of cutoff and forward active so

$$V_{DE} = V_B = 0.5V.$$

So,

$$\frac{V_{IH} - V_B}{5} = \frac{V_B + 5}{100}$$

∴ $V_{IL} = 0.775V$.

Determination of V_{IH} ,

Transistor is on the verge of forward active and saturation so

$$\beta_F = 30, V_{BE} = 0.8V, V_{CE} = 0.2V$$

$$I_{RB} = \frac{0.8 + 5}{100} = 0.058mA$$

$$I_C = \frac{5 - 0.2}{2} = 2.4mA$$

$$\beta_F = \frac{I_C}{I_B}$$

∴ $I_B = 0.08mA$

$$\therefore I_1 = I_{RB} + I_B = 0.058 + 0.08 = 0.138mA$$

Now,

$$\frac{V_{IH} - 0.8}{5} = 0.138$$

∴ $V_{IH} = 1.49V$.

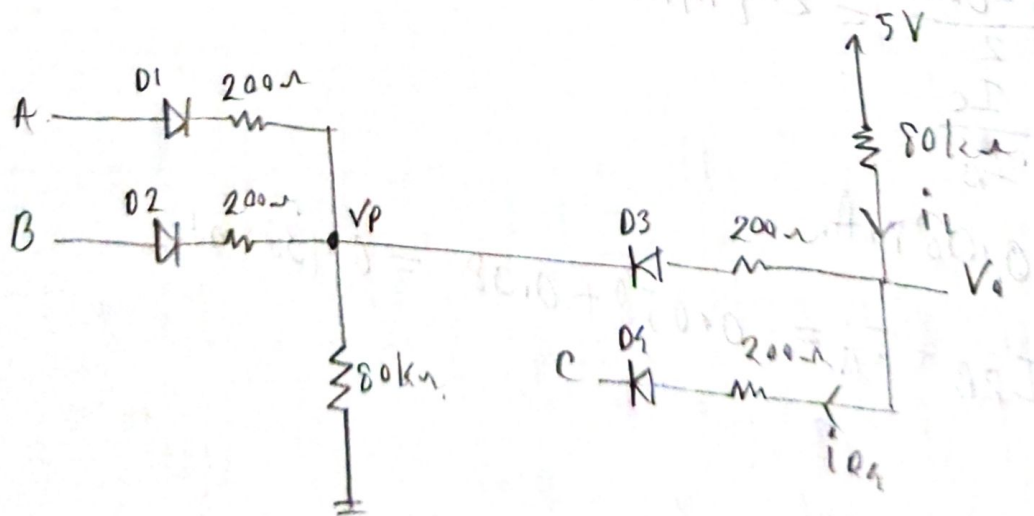
$$\begin{aligned} \text{So, } V_{NL} &= V_{iL} - V_{oh} \\ &= (0.775 - 0.2) \text{ V} \\ &= 0.575 \text{ V} \end{aligned}$$

$$\begin{aligned} V_{NH} &= V_{oh} - V_{iH} \\ &= (4.5 - 1.49) \text{ V} \\ &= 3.01 \text{ V} \end{aligned}$$

$$\begin{aligned} \therefore V_{NM} &= \min(V_{NL}, V_{NH}) \\ &= \min(0.575, 3.01) \\ &= 0.575 \text{ V} \end{aligned}$$

Ans

Ans to the Q No-4



(a) The circuit implementation = $(A+B) \cdot C$ Ans

(b) As per question, $A = \text{low (0V)}$, $B = \text{high (5V)}$, $C = \text{low (0V)}$

If $A=0$ and $B=5V$ so $D1 = \text{off}$ and $D2 = \text{ON}$ meaning the output along V_p point is high. That makes $D3 = \text{off}$

So,

$$i_1 = i_4$$

$$\frac{5 - V_o}{80 \times 10^3} = \frac{V_o - 0.7}{200}$$

$$\Rightarrow 5 - V_o = 400 V_o - 280$$

$$\Rightarrow 401 V_o = 285$$

$$\Rightarrow V_o = 0.71 V$$

$$\therefore V_o = 0.71 V$$

$$\therefore \text{Power dissipation} = (5-0) \times \frac{5-0.7-0}{200+80 \times 10^3} + (5-0) \times \frac{5-0.71}{80 \times 10^3}$$

$$= 0.536 \text{ mW}$$

Ans